ISS 2103 • EXAM REVISION PACK 05

Network Layer: IPv4, Subnetting/VLSM, IPv6 & DHCP

Course outline weeks 5–7: internetworking, the IP packet, IP addressing, IPv6

Priority: **VERY HIGH**

A subnetting calculation has appeared in **every** paper (4–12 marks). This is the topic that wins or loses the exam. The marks are mostly for the *working*, so this pack shows every step and why you do it.

How often this topic is examined

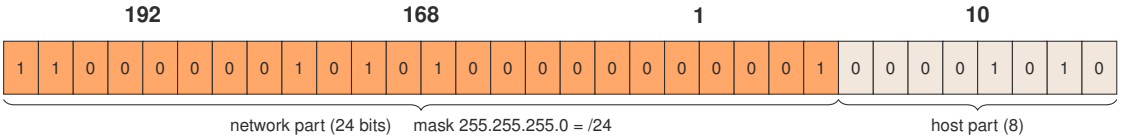
Question asked	Paper(s)	Marks
Subnetting calculation (subnets, hosts, ranges, broadcast)	2018 Q3a, 2023 Q1d, Q3c, ISA2024 Q1d, 2025 Q4a	4–12
IPv4 header: draw it + purpose of fields	2023 Q1a, ISA2024 Q1a	6–7
IPv6 header diagram	2018 Q1d	5
Advantages of IPv6 over IPv4	2023 Q1b	5
Shorten IPv6 addresses	2023 Q5e, ISA2024 Q3a	1–2
SLAAC vs DHCPv6	2025 Q3b, ISA2024 Q5d	6–8
Purpose & benefits of subnetting	2025 Q4b, ISA2024 Q2b	5–6
Network segmentation	ISA2024 Q1b	6
Define ARP	2018 Q3b(iii)	3
DHCP four-step (DORA) with diagrams	2018 Q4a	10
Static vs dynamic IP	SE2024 Q1b(iv)	2

EXAM TIP

Practise the method in Section 2 until a full subnetting question takes you under 10 minutes. Always write the binary mask, the block size and a results table.

1 IPv4 addressing basics

An IPv4 address is a **32-bit logical address** written as four 8-bit octets in dotted decimal. It has a **network part** (which network) and a **host part** (which device on it). The **subnet mask** marks network bits with 1s and host bits with 0s.



Each octet is 8 bits, so it can only be 0–255. If you see 256 in an address, it is wrong.

Class	1st octet	Lead bits	Default mask	Hosts/net	Private range
A	1–126	0	255.0.0.0 (/8)	16 777 214	10.0.0.0 – 10.255.255.255
B	128–191	10	255.255.0.0 (/16)	65 534	172.16.0.0 – 172.31.255.255
C	192–223	110	255.255.255.0 (/24)	254	192.168.0.0 – 192.168.255.255
D	224–239	1110	–	multicast	–
E	240–255	1111	–	reserved	–

- **Network address** = all host bits 0. **Broadcast address** = all host bits 1. Neither can be given to a host, so usable hosts = $2^h - 2$.
- **Classless (CIDR)**: no fixed classes; the prefix /n says how many bits are network, e.g. 14.24.74.0/24. Gives efficient allocation and route aggregation (supernetting).
- **NAT**: a router translates private addresses to one public address, so many hosts share it. Saves IPv4 addresses and hides internal hosts.

127.x.x.x is **loopback** (127.0.0.1 = this computer). That is why class A stops at 126.

PAST PAPER **SE2024 Q1b(iv)** **[2 marks]**

Differentiate between a static and a dynamic IP address.

Model answer

Static IP	Dynamic IP
Configured manually; does not change	Assigned automatically by a DHCP server for a lease; may change
Used for servers, printers, routers (things others must find)	Used for client PCs and phones
More admin work; risk of conflicts	Easy to manage; addresses reused efficiently

PAST PAPER **2025 Q4b / ISA2024 Q2b** **[5–6 marks]**

Explain the purpose of subnetting and the benefits it provides to network administrators. / Discuss the concept of IP addressing and subnetting.

Model answer

Subnetting divides one large network into smaller logical networks (subnets) by **borrowing bits from the host part**, giving a longer subnet mask.

1. **Less broadcast traffic**: each subnet is its own broadcast domain, so performance improves and congestion drops.
2. **Efficient use of addresses**: each department gets the right-sized block (VLSM), so less is wasted.
3. **Security**: traffic between subnets passes through a router or firewall where ACLs apply; sensitive departments are isolated.
4. **Easier management and troubleshooting**: a fault stays inside one subnet; the network is organised by department, floor or site.
5. **Scalability**: new subnets are easy to add; route summarisation keeps routing tables small.

PAST PAPER **ISA2024 Q1b** **[6 marks]**

Explain the concept of network segmentation.

Model answer

Network segmentation splits one flat network into smaller, isolated segments, each with its

own broadcast domain and its own security rules. Traffic between segments must pass a controlled point (router, layer-3 switch or firewall).

- **How**: physically (separate switches/routers), at layer 2 with **VLANs**, at layer 3 with **IP subnets**, or with firewalls/ACLs (micro-segmentation in data centres).
- **Performance**: smaller broadcast domains mean less broadcast traffic and fewer collisions.
- **Security**: an attacker or worm that gets into one segment cannot freely reach the others; sensitive systems (servers, finance, CCTV) sit in their own zone, e.g. a **DMZ** for public servers.
- **Containment and troubleshooting**: faults stay inside one segment; easier monitoring.
- **Compliance**: e.g. card-payment systems kept separate (PCI DSS).

Example: a university puts Staff (VLAN 10, 10.1.10.0/24), Students (VLAN 20, 10.1.20.0/24) and Servers (VLAN 30) in separate segments; a firewall only lets students reach the web server, not the finance server.

Segmentation is the **security** word for subnetting/VLANs. In an ISA exam, always mention containing attacks.

2 The subnetting method (learn this)

Bits borrowed in the octet	1	2	3	4	5	6	7	8
Mask octet value	128	192	224	240	248	252	254	255
Block size = 256 − mask	128	64	32	16	8	4	2	1

/24	/25	/26	/27	/28	/29	/30
254	126	62	30	14	6	2

usable hosts per prefix (last octet)

The five formulas

$$s = \text{new prefix} - \text{default prefix} \quad \leftarrow \textit{bits borrowed}$$
$$\text{subnets} = 2^{s}$$
$$h = 32 - \text{prefix} \quad \leftarrow \textit{host bits left}$$
$$\text{usable hosts} = 2^{h} - 2 \quad \leftarrow \textit{minus network \& broadcast}$$
$$\text{block size} = 256 - \text{mask value in the interesting octet} \quad \leftarrow \textit{the octet that is not 255 or 0}$$

REMEMBER IT

Subnets count up in block sizes. **Network** = a multiple of the block. **Broadcast** = next network −1. **First host** = network +1. **Last host** = broadcast −1.

TRAP – where students lose marks

- Using 2^{h} instead of $2^{h} - 2$ for usable hosts.
- Putting the block size in the wrong octet. For a /22 the interesting octet is the **third**, so the 4th octet of the broadcast is **255**, not 3.
- Counting borrowed bits from /0 instead of from the class default.

3 Worked past-paper calculations

PAST PAPER **2025 Q4a** **[12 marks]**

Given the network 172.16.0.0 with a subnet mask of 255.255.252.0: (i) which class? (ii) number of possible subnets (iii) usable hosts per subnet (iv) network & broadcast addresses for the first two subnets (v) valid host ranges for the first three subnets.

Model answer

Working

$$255.255.252.0 = 11111111.11111111.\underbrace{111111}_{s=6}\underbrace{00.00000000}_{h=10} = /22$$ ← *count the 1s*

(i) $172 \in 128\text{–}191 \Rightarrow$ **Class B** ← *default /16*

(ii) $s = 22 - 16 = 6$, $2^6 = 64$ subnets

(iii) $h = 32 - 22 = 10$, $2^{10} - 2 = 1022$ hosts ← $1024 - 2$

block $= 256 - 252 = 4$ ← *in the 3rd octet!*

Subnet	Network	First host	Last host	Broadcast
1	172.16.0.0	172.16.0.1	172.16.3.254	172.16.3.255
2	172.16.4.0	172.16.4.1	172.16.7.254	172.16.7.255
3	172.16.8.0	172.16.8.1	172.16.11.254	172.16.11.255

Third octet goes 0, 4, 8, 12, … Each subnet spans 4 values of the 3rd octet × 256 = 1024 addresses.

(iv) Subnet 1: 172.16.0.0 / 172.16.3.255. Subnet 2: 172.16.4.0 / 172.16.7.255.
(v) 172.16.0.1–172.16.3.254; 172.16.4.1–172.16.7.254; 172.16.8.1–172.16.11.254.

PAST PAPER **ISA2024 Q1d** **[6 marks]**

You have sub-netted your class C network 192.168.1.0 with a subnet mask of 255.255.255.240. Calculate (i) number of networks (ii) hosts per network (iii) the full range of the first three networks (iv) the usable address ranges of those networks.

Model answer

Working

$240 = 11110000 \Rightarrow /28$ ← *4 borrowed, 4 host bits*

(i) networks $= 2^4 = 16$

(ii) hosts $= 2^4 - 2 = 14$

block $= 256 - 240 = 16$ ← *networks at .0, .16, .32 …*

Net	Full range (network – broadcast)	Usable hosts
1	192.168.1.0 – 192.168.1.15	192.168.1.1 – 192.168.1.14
2	192.168.1.16 – 192.168.1.31	192.168.1.17 – 192.168.1.30
3	192.168.1.32 – 192.168.1.47	192.168.1.33 – 192.168.1.46

PAST PAPER **2023 Q3c** **[4 marks]**

A host was given the IP address 198.168.4.121/27. Indicate (i) the network address (ii) the broadcast address (iii) the total number of hosts available (iv) whether 192.168.4.93 can be the default gateway of the host. Explain.

Model answer

Working

$$/27 \Rightarrow 255.255.255.224, \quad \text{block} = 256 - 224 = 32$$

networks : 0, 32, 64, 96, 128, . . . ← *121 lies in 96–127*

(i) network = 198.168.4.96

(ii) broadcast = $128 - 1 \Rightarrow$ 198.168.4.127 ← *next network −1*

(iii) hosts = $2^5 - 2 = 30$ ← *.97 – .126*

(iv) **No.** A default gateway must be in the **same subnet** as the host. 192.168.4.93 has a different network part (192 vs 198), and even its last octet (.93) falls in the .64–.95 subnet, not .96–.127. A valid gateway would be e.g. 198.168.4.97.

The paper prints **198**.168 for the host but **192**.168 for the gateway, which may be a typo. Either way the answer is **No**: .93 is in the .64–.95 subnet, not .96–.127.

3.1 VLSM: different-sized subnets

REMEMBER IT

VLSM rule: sort requirements **largest first**; give each the smallest block where $2^h - 2 \geq$ hosts needed; allocate blocks one after the other.

PAST PAPER **2023 Q1d** **[6 marks]**

Considering IPv4 classless addressing, an organisation is granted the block 14.24.74.0/24. It needs 3 subnets: one of 120 addresses, one of 60 and one of 10. Show the network address and subnet mask of each subnet using VLSM. Show all working.

Model answer

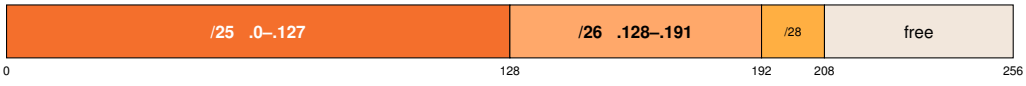
Working

$$120 + 2 \leq 128 = 2^7 \Rightarrow h = 7 \Rightarrow /25 \ (255.255.255.128) \quad \leftarrow 2^6 = 64 \text{ is too small}$$

$$60 + 2 \leq 64 = 2^6 \Rightarrow h = 6 \Rightarrow /26 \ (255.255.255.192)$$

$$10 + 2 \leq 16 = 2^4 \Rightarrow h = 4 \Rightarrow /28 \ (255.255.255.240) \quad \leftarrow 2^3 = 8 \text{ too small}$$

Need	Network	Mask	Usable hosts	Broadcast
120	14.24.74.0/25	255.255.255.128	.1 – .126	.127
60	14.24.74.128/26	255.255.255.192	.129 – .190	.191
10	14.24.74.192/28	255.255.255.240	.193 – .206	.207



The /24 address space after VLSM: 48 addresses (.208–.255) left for growth.

PAST PAPER **2018 Q3a** **[10 marks]**

Company ABC has: Accounts 10, HR 4, IT 4, Sales & Marketing 20 employees. Using 192.168.0.0/24, design a subnetting scheme that does not waste addresses and caters for an additional two members in each subnet.

Model answer

Working

$$\text{Sales } 20+2=22 \Rightarrow 2^5-2=30 \Rightarrow /27 \quad \leftarrow \textit{biggest first}$$
$$\text{Accounts } 10+2=12 \Rightarrow 2^4-2=14 \Rightarrow /28$$
$$\text{HR } 4+2=6 \Rightarrow 2^3-2=6 \Rightarrow /29 \quad \leftarrow \textit{exactly fits}$$
$$\text{IT } 4+2=6 \Rightarrow 2^3-2=6 \Rightarrow /29$$

Dept	Network	Mask	Usable hosts	Broadcast
Sales	192.168.0.0/27	255.255.255.224	.1 – .30	.31
Accounts	192.168.0.32/28	255.255.255.240	.33 – .46	.47
HR	192.168.0.48/29	255.255.255.248	.49 – .54	.55
IT	192.168.0.56/29	255.255.255.248	.57 – .62	.63

Total used: 64 addresses; 192.168.0.64–.255 is free for future subnets.

State your assumption. If each subnet also needs a router/gateway address, HR and IT need 7 → /28 each (HR .48/28, IT .64/28).

4 The IPv4 header

PAST PAPER **ISA2024 Q1a / 2023 Q1a** **[6–7 marks]**

Draw the typical structure of an IP header and state the purpose of: Version, Differentiated Services, Time to Live, Protocol, Source and Destination IP addresses.

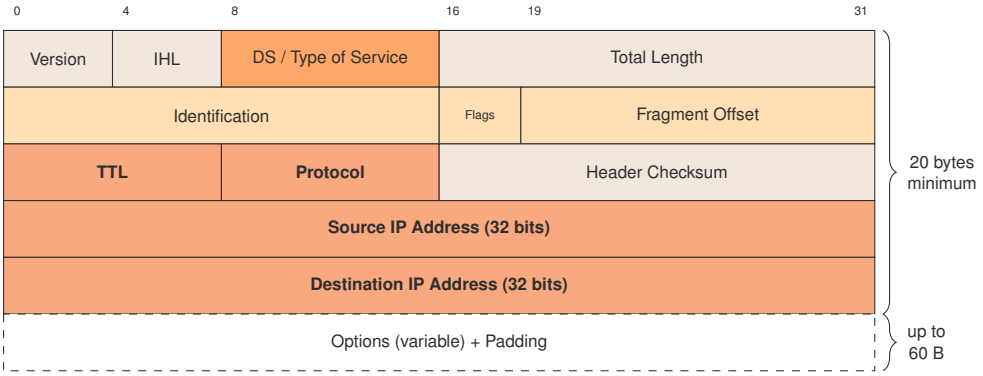


Figure: IPv4 header, 32 bits wide. Orange = fields the exam asks about; yellow = fragmentation.

Model answer

Field	Purpose
Version (4)	IP version, 4 for IPv4, so the router knows how to read the header.
IHL (4)	Header length in 32-bit words (5 = 20 bytes, max 15 = 60 bytes).
Differentiated Services (8)	Priority / class of service for QoS (DSCP 6 bits + ECN 2 bits), e.g. VoIP marked high priority.
Total Length (16)	Length of the whole datagram (header + data), max 65 535 bytes.
Identification, Flags, Offset	Fragmentation and reassembly: ID groups fragments; flags DF / MF; offset gives position.
Time to Live (8)	Max number of hops. Each router subtracts 1; at 0 the packet is discarded and ICMP Time Exceeded is sent. **Stops packets looping forever.**
Protocol (8)	Upper-layer protocol the data belongs to: 1 = ICMP, 6 = TCP, 17 = UDP, 89 = OSPF.
Header Checksum (16)	Error detection on the header only; recalculated at each router.
Source IP (32)	Logical address of the original sender; used to reply.
Destination IP (32)	Logical address of the final receiver; routers use it to forward.
Options + Padding	Optional (security, record route, timestamp); padding keeps a multiple of 32 bits.

5 IPv6

Diagram + 1 line per field. The diagram alone is usually worth about half the marks.

IPv6 uses **128-bit** addresses: 8 groups of 4 hex digits separated by colons, e.g. 2001:0DB8:0000:1470:0000:0000:0000:0200. Types: unicast (global 2000::/3, link-local FE80::/10, unique local FC00::/7), multicast (FF00::/8), anycast. **There is no broadcast.**

Address space

$2^{128} \approx 3.4 \times 10^{38}$ addresses ← *vs* $2^{32} \approx 4.3 \times 10^{9}$ *in IPv4*

5.1 Shortening rules

1. Remove **leading** zeros in any group (0DB8 → DB8, 0000 → 0, 0200 → 200).
2. Replace **one** run of consecutive all-zero groups with `::`. Only once per address; use it on the longest run.

PAST PAPER **2023 Q5e / ISA2024 Q3a** **[1–2 marks]**

Express in shortened form: (i) 2001:0DB8:0000:1470:0000:0000:0000:0200 (ii) F380:0000:0000:0000:0123:4567:89AB:CDEF (iii) 2001:0DB8:0000:1470:0000:0000:0000:0300

Model answer

Working

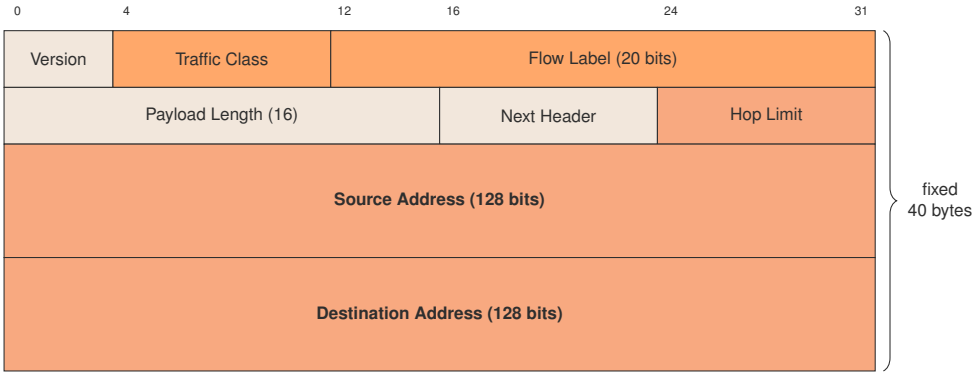
(i) 2001:DB8:0:1470:0:0:0:200 ← *rule 1*
→ **2001:DB8:0:1470::200** ← *rule 2 on the run of 3 zeros*
(ii) F380:0:0:0:123:4567:89AB:CDEF → **F380::123:4567:89AB:CDEF**
(iii) **2001:DB8:0:1470::300**

TRAP – where students lose marks

2001:DB8::1470::300 is **wrong**. With two `::` the receiver cannot tell how many zeros each one hides. Also never drop *trailing* zeros: 0200 → 200, not 2.

PAST PAPER **2018 Q1d** **[5 marks]**

Draw a fully labelled diagram of the IPv6 header.



No checksum, no fragmentation fields, no options (extension headers instead).

- **Version** = 6. **Traffic class** = priority/QoS (like IPv4 DS). **Flow label** = marks packets of one flow needing special handling (real-time audio/video).
- **Payload length** = bytes after the 40-byte header. **Next header** = type of the next header (like IPv4 Protocol). **Hop limit** = like TTL.

PAST PAPER **2023 Q1b** **[5 marks]**

Indicate five advantages of IPv6 over IPv4.

Model answer

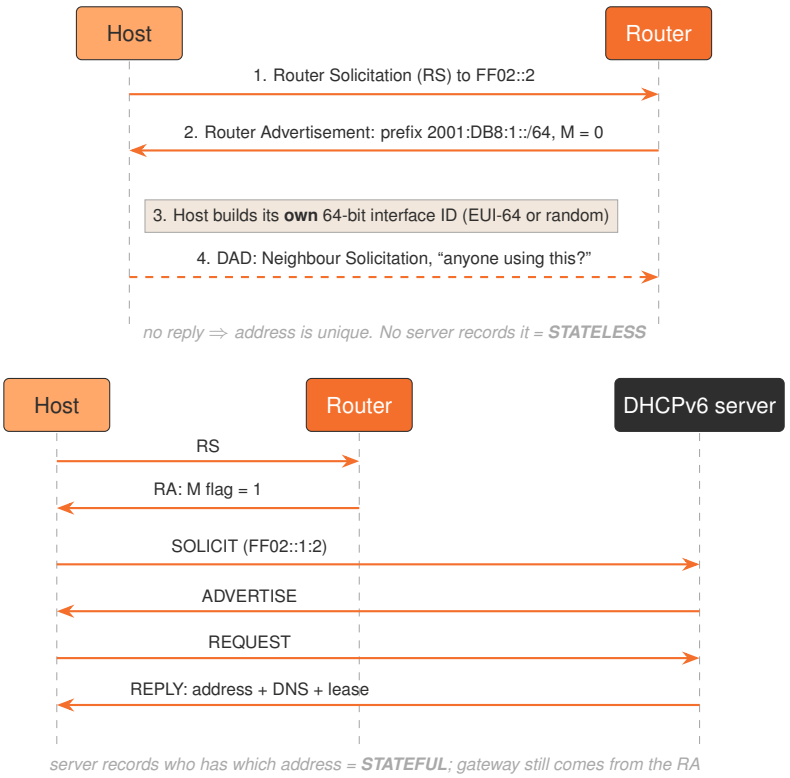
1. **Huge address space**: 128 bits (3.4×10^{38}) ends IPv4 exhaustion; no need for NAT, so true end-to-end connectivity.
2. **Simpler fixed 40-byte header**: fewer fields, no checksum, no router fragmentation, so faster processing.
3. **Built-in security**: IPsec (authentication and encryption) is part of the design.
4. **Auto-configuration**: SLAAC gives plug-and-play addressing; DHCPv6 is also available.
5. **Better QoS**: flow label and traffic class for real-time traffic.

Extra: efficient multicast/anycast and no broadcast storms; Mobile IPv6; extension headers; hierarchical addressing gives smaller routing tables.

6 SLAAC vs DHCPv6

PAST PAPER **2025 Q3b / ISA2024 Q5d** **[6–8 marks]**

Global unicast IPv6 addresses can be assigned dynamically using SLAAC and DHCPv6. Compare and contrast them, explaining the key differences in how they assign IPv6 addresses.



Model answer

Point	SLAAC	DHCPv6 (stateful)
Who makes the address	The host: /64 prefix from the router's RA + its own 64-bit interface ID (EUI-64 or random)	A DHCPv6 server assigns the whole address from its pool
Server needed?	No, only a router sending RAs	Yes, a DHCPv6 server (or a router acting as one)
State	Stateless: nobody records who has which address	Stateful: lease database, good for auditing
Messages	ICMPv6 RS / RA; DAD with Neighbour Solicitation	Solicit, Advertise, Request, Reply (UDP 546/547)
DNS etc.	Limited: RDNSS option, or stateless DHCPv6 (O = 1)	Gives DNS, domain name and other options
RA flags	M = 0	M = 1 (managed)
Control	Simple, plug-and-play, scales easily	More admin control; easier to manage and secure

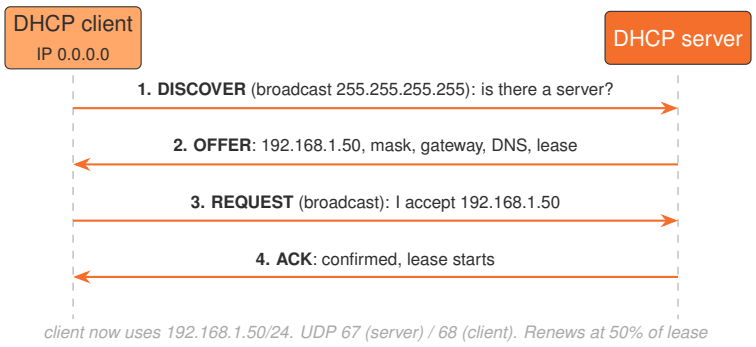
Similarities: both are dynamic, both rely on the RA, both give global unicast addresses, and in both the gateway comes from the RA.

"Compare **and contrast**" means list a similarity too. One line of similarities earns easy marks.

7 DHCP (IPv4): the DORA process

PAST PAPER **2018 Q4a** **[10 marks]**

Describe the DHCP four-step process with the aid of diagrams.



Model answer

DHCP automatically gives hosts an IP address, subnet mask, default gateway, DNS server and lease time.

1. **Discover**: the new client broadcasts DHCPDISCOVER (source 0.0.0.0, destination 255.255.255.255) to find a server.
2. **Offer**: each server replies with DHCPOFFER: an available IP, mask, gateway, DNS and lease time.
3. **Request**: the client picks one offer and broadcasts DHCPREQUEST, so other servers know they were declined.
4. **Acknowledge**: the chosen server sends DHCPACK and the client configures itself. If the address is gone, it gets DHCPNAK and starts again.

Extra marks: renewal at 50% (unicast REQUEST), rebinding at 87.5%; DHCPRELEASE on shutdown; a relay agent (`ip helper-address`) forwards broadcasts to a server on another subnet.

REMEMBER IT

DORA: Discover, Offer, Request, Acknowledge. Client **68**, server **67**.

8 ARP: from IP address to MAC address

PAST PAPER **2018 Q3b(iii)** **[3 marks]**

Define ARP.



Model answer

ARP (Address Resolution Protocol) finds the **MAC (physical) address** that belongs to a known **IPv4 address** on the local network. The sender broadcasts an ARP request; the owner of that IP replies by unicast with its MAC; the result is cached in the ARP table. It links layer 3 to layer 2, because a frame needs a destination MAC. (In IPv6, Neighbour Discovery does this job.)

If the destination is on **another** network, the host ARPs for the **default gateway's** MAC, not the remote host's.

9 Practice questions

PRACTICE Q1 EASY [2 marks]

How many usable hosts does a /26 subnet give? Show the formula.

Answer

$h = 32 - 26 = 6$, $2^6 - 2 = 62$ hosts.

PRACTICE Q2 EASY [3 marks]

State the class and default mask of: (a) 10.4.5.6 (b) 150.1.1.1 (c) 200.9.9.9.

Answer

(a) Class A, 255.0.0.0 (b) Class B, 255.255.0.0 (c) Class C, 255.255.255.0.

PRACTICE Q3 EASY [2 marks]

Shorten FE80:0000:0000:0000:0202:B3FF:FE1E:8329.

Answer

FE80::202:B3FF:FE1E:8329

PRACTICE Q4 MEDIUM [5 marks]

A host has address 172.20.45.200/20. Find the network address, broadcast address and the number of usable hosts.

Answer

Working

$$
\begin{aligned}
/20 &\Rightarrow 255.255.240.0 && \leftarrow \textit{3rd octet is interesting} \\
\text{block} &= 256 - 240 = 16 && \leftarrow \textit{3rd octet: 0, 16, 32, 48 \ldots} \\
45 &\in [32, 47] \Rightarrow \text{network } 172.20.32.0 \\
\text{broadcast} &= 172.20.(48 - 1).255 = 172.20.47.255 \\
\text{hosts} &= 2^{12} - 2 = 4094
\end{aligned}
$$

PRACTICE Q5 MEDIUM [6 marks]

Explain three differences between the IPv4 and IPv6 headers.

Answer

(1) IPv4 is 20–60 bytes and variable; IPv6 is a fixed 40 bytes. (2) IPv4 has a header checksum; IPv6 has none (left to layers 2 and 4). (3) IPv4 routers can fragment (ID/flags/offset); in IPv6 only the source fragments, using an extension header. Also: TTL $\rightarrow$ Hop Limit, Protocol $\rightarrow$ Next Header, and IPv6 adds a Flow Label.

PRACTICE Q6 HARD [12 marks]

You are given 192.168.10.0/24. Design a VLSM scheme for: Lab 100 hosts, Admin 50, Library 25, two point-to-point WAN links of 2 hosts each. Give network, mask, usable range and broadcast for each.

Answer

Working

$$100 \to 2^7 - 2 = 126 \Rightarrow /25 \qquad \leftarrow \textit{largest first}$$
$$50 \to 2^6 - 2 = 62 \Rightarrow /26$$
$$25 \to 2^5 - 2 = 30 \Rightarrow /27$$
$$2 \to 2^2 - 2 = 2 \Rightarrow /30 \qquad \leftarrow \textit{the classic WAN-link mask}$$

Subnet	Network	Mask	Usable	Bcast
Lab	.0/25	255.255.255.128	.1–.126	.127
Admin	.128/26	255.255.255.192	.129–.190	.191
Library	.192/27	255.255.255.224	.193–.222	.223
WAN 1	.224/30	255.255.255.252	.225–.226	.227
WAN 2	.228/30	255.255.255.252	.229–.230	.231

Free: 192.168.10.232–.255 for growth.

PRACTICE Q7 HARD **[10 marks]**

A class B network 150.50.0.0 must be split into at least 500 subnets. Find the mask, the number of subnets, the hosts per subnet and the first two subnet ranges.

Answer

Working

$$2^s \geq 500 \Rightarrow s = 9\ (2^9 = 512) \qquad \leftarrow 2^8 = 256 \textit{ is too few}$$
$$\text{prefix} = 16 + 9 = /25 \Rightarrow 255.255.255.128$$
$$h = 32 - 25 = 7 \Rightarrow 2^7 - 2 = 126 \text{ hosts}$$
$$\text{block} = 128 \text{ in the 4th octet} \qquad \leftarrow \textit{borrowed bits cross into octet 4}$$

Subnet 1: 150.50.0.0 – hosts .0.1–.0.126 – broadcast 150.50.0.127.
Subnet 2: 150.50.0.128 – hosts .0.129–.0.254 – broadcast 150.50.0.255.

ONE-PAGE CHEAT SHEET

Formulas

- Subnets = 2^s, hosts = $2^h - 2$
- Block = 256− mask octet
- Broadcast = next network −1
- /25 128 /26 64 /27 32 /28 16 /29 8 /30 4
- Hosts: /24 254, /25 126, /26 62, /27 30, /28 14, /29 6, /30 2

Classes

- A 1–126 /8, B 128–191 /16, C 192–223 /24
- Private: 10/8, 172.16–31, 192.168/16
- 127 = loopback

VLSM: biggest first, smallest block where $2^h - 2 \geq$ need, state assumptions.

DHCPv6: server assigns and records, stateful, M = 1, Solicit/Advertise/Request/Reply.
DHCP DORA: Discover, Offer, Request, Ack. UDP 67 server / 68 client. Renew 50%.

IPv4 header (20–60 B): Version, IHL, DS, Total length, ID/Flags/Offset, **TTL**, **Protocol** (1 ICMP, 6 TCP, 17 UDP), Checksum, **Src**, **Dst**.
IPv6 header (fixed 40 B): Version, Traffic class, Flow label, Payload length, Next header, Hop limit, Src (128), Dst (128).
IPv6 shorten: drop leading zeros; `::` once, on the longest zero run.
SLAAC: host builds address from RA prefix + own ID, stateless, M = 0.